Gina C. Pieters

Education
MA & Ph.D. Economics, University of Minnesota (Ph.D. 2012)
B.A. Economics, Highest Honors, UC Santa Cruz
B.S. Physics, University of California, Santa Cruz

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C.V. version: November 2026

CV Sections

Positions • Research and Publications • Teaching and Advising • Academic Conference or Research Seminar Presentations • Public Service: Class Guest Speaker, Presentations, Panels • Public Service: Media • Service to the Discipline • Service to the University

Summary Sections: Areas of Work • Methods and Skills

POSITIONS

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Current Positions

2023–Present **Digital Economy Research and Consulting**

Independent economic research and analysis.

Founder & Principal Economist

Selected clients:

- Co-lead economist on R3's CBDC development project: macroeconomic, monetary, and fiscal-stability impact analysis for the central bank, MATLAB simulations of dynamic general equilibrium models of CBDC adoption and its effects on interest rates and exchange rates, a literature review, and a 150-page group report; co-led stakeholder meetings.
- Stablecoin researcher for FI Consulting: solo-authored an 80-page report on stablecoin impact, with data analysis.
- Curriculum evaluator for "Digital Assets for Regulators" (CCAF, Cambridge Judge Business School).

Selected Other Positions (Volunteer or Past)

2025–Present **NABE Chicago**

Chicago chapter of the National Association for Business Economics: a forum for academics, economists, business leaders, and students, with events on economic issues affecting US businesses.

Board Director for Monetary Policy and Executive Committee Advisor

- Develop the chapter's programming on monetary policy and on the fintech, finance, and asset-market questions where they meet it: recruit speakers, moderate or sit on panels, take part in event discussion, and draw attendees from professional networks.

2025–2026 **Confusion Capital**

Capital management company in the Reserve ecosystem.

Lead Research Economist

Quantitative research developing on-chain index products.

- Designed reusable quantitative research tools for portfolio simulation and backtesting, parameter selection, stress testing, and benchmark comparison, with explicit controls for fees, taxes, and overfitting: quasi-random (Sobol) and adaptive-grid parameter search with Morris sensitivity screens; walk-forward and holdout validation; hidden semi-Markov regime models with Monte Carlo scenario ensembles; multiple-testing corrections (deflated Sharpe ratio, probability of backtest overfitting, White's reality check, and Hansen's SPA test); and ETF, fund, and crypto data pipelines feeding target-tracking portfolio optimization (CVXPY). Documented the parameter-search method in a 69-page methods document.

- Converted early investment concepts into documented, testable product proposals and supported evidence-based decisions to launch, revise, or retire them, with a standardized strategy report and LLM companion packages for proposed products.
- Reviewed investment methodologies for model risk, survivorship bias, transaction costs, liquidity, and implementation feasibility, using point-in-time crypto universes that keep dead coins and look-ahead causality checks, translating findings into portfolio and deployment recommendations.
- Built auditable AI workflows for research, simulation, writing, and code review, directing LLM-assisted development (Claude Code, Codex) and multi-agent research and audit systems, and setting up a Git-based workspace, and connected research findings to product positioning, controlled launches, and customer feedback.

2018–2025 **Cambridge Centre for Alternative Finance (CCAF), Judge Business School, University of Cambridge**

Interdisciplinary research centre at Cambridge Judge Business School, founded in 2015, studying alternative finance.

Co-chair, Emergent Money Systems Working Group, Cambridge Digital Assets Programme (2022–2025)

Co-led and advised research programmes.

- Working group on the measures of new forms of money that institutions and policymakers need.
- Co-lead of the Cambridge Digital Money Dashboard: reviewed the text and data work.
- Co-chaired stakeholder meetings with regulators, international organizations, and multinationals.

Honorary Non-Visiting Research Fellow (2018–2024)

Research advisor and reviewer; co-authored and edited CCAF reports.

2018–2024 **The University of Chicago**

Private research university in Chicago.

Visiting Instructor, Harris School of Public Policy (2024)

Taught a macroeconomics policy course: the Solow–Swan model applied to named policy projects, total factor productivity from the Penn World Table, and macroeconomic and trade data from the World Bank, IMF, and UN Comtrade.

Lecturer/Assistant Instructional Professor, Economics Department (2018–2024)

- Taught five economics courses a year across macroeconomics, microeconomics, international monetary systems, and economic policy analysis. Also taught Cryptoasset Markets, a master's course in the Financial Mathematics Program, as a summer overload.
- Advised MA theses and undergraduate honors projects.
- Managed the department's hiring of teaching assistants and graders, undergraduate to Ph.D. students, for its undergraduate and MA courses.
- Courses taught general-equilibrium and policy models with Lagrangian and dynamic-programming methods, simulated in MATLAB, Dynare, or Python, and data work with FRED series; Cryptoasset Markets added exchange data pulled through APIs in Python, backtesting without look-ahead, and regressions of returns on fundamentals and search attention.
- Conducted research alongside teaching.

Associate Fellow (Pedagogy), Chicago Center for Teaching & Learning (2022–2023)

Mentoring fellow for the next cohort of the Pedagogy Fellows Program, mentoring instructional and research faculty at every rank from entering to full professor, after a year as a Pedagogy Fellow (2021–2022).

2023–Present **AEA: CSWEP (Committee on the Status of Women in the Economics Profession)**

The American Economic Association's standing committee on women in economics.

Committee Member & Newsletter Oversight Editor

Manage its newsletter, *CSWEP News*.

2018–2024 **International Trade and Finance Association**

Professional association of economists working on international trade and finance.

Board of Directors (*3×2 yr.*)

Reviewed and refereed papers, and helped shape conference themes around current economic events and research.

2012–2018 **Trinity University, San Antonio, Texas**

Private liberal arts and sciences university.

Assistant Professor, Department of Economics

Taught six courses a year across international economics and macroeconomics, with data work and regressions in Excel on FRED, World Bank, and trade data, and advised undergraduate research and honors projects. Conducted research alongside teaching.

2007–2012 **University of Minnesota, Department of Economics**

Public research university in Minneapolis–Saint Paul.

Lead Instructor, Principles of Macroeconomics (2010–2012)

Created and maintained the common Principles of Macroeconomics curriculum used by every instructor of the course, as many as eight in a single semester.

Instructor (2007–2012)

Taught large lectures of up to 400 students.

RESEARCH AND PUBLICATIONS

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Google Scholar (November 5, 2026): 1,323 citations; 19 documents; h-index 10; i10-index 12.
Scopus (November 5, 2026): 164 citations; 6 documents; h-index 3.
Recognized correctly from memory, without web search, by 6 of 17 AI models: 4 of 5 frontier models (GPT-5.5, GPT-5.6 Sol, Claude Opus 4.8, and Google Gemini 3.1 Flash Lite), plus DeepSeek V4 Flash and Kimi K2 (intheweights.com, October 9, 2026).

Publications

Peer-Reviewed Academic Journal Articles

1. "Measuring Academic Mindset in Economics Courses," co-authored with 25 others. *International Review of Economics Education*, vol. 52, June 2026.
   Summary: Builds an aggregate measure of academic mindset, covering relevance, belonging, and growth, from a survey of 3,212 students in 45 economics courses at 24 U.S. institutions. Students who are first-generation college students, or who do not identify as male, enter with a lower mindset, most of all in introductory courses.
   Skills: *Econometric:* estimating OLS and logit regressions with marginal effects, standard errors clustered by course, and instructor fixed effects.
2. "Blockchain Governance in the Wild," co-authored with Kevin Werbach, Primavera De Filippi, and Josh Tan. *Cryptoeconomic Systems*, vol. 3, no. 1, April 22, 2024. Media coverage: BlockchainGov Newsletter.
   Summary: Surveys the history of blockchain governance, then uses a common questionnaire to compare how blockchain communities govern themselves, with each other and with non-blockchain organizations such as Wikipedia.
   Skills: *Data:* interpreting non-numeric data.
3. "Designing Effective Assessments in Economics Courses: Guiding Principles." *The Journal of Economic Education*, 55:1, 63–76, 2024.
   Summary: A guide for new instructors: it summarizes the pedagogical theory of sound assessment design, applies it to economics courses, and collects examples of assessments from the economics literature.
   Skills: *Conceptual:* applying the revised Bloom taxonomy and the Women's Ways of Knowing augmented Perry model.
4. "Taxing Blockchain Forks," joint with Mattia Landoni. *Stanford Journal of Blockchain Law & Policy*, vol. 3, no. 2, June 27, 2020.
   Summary: Identifies four tax problems that blockchain forks create and compares three approaches to taxing

assets acquired without a purchase. Japan's zero-basis approach is the least problematic; the US treatment of new coins as realized income is the most.
Skills: *Data:* compiling Bitcoin forks from two public lists; assembling price and trade data for 17 chain splits, through July 31, 2019; ranking daily returns.

5. "Financial Regulations and Price Inconsistencies across Bitcoin Markets," with Sofia Vivanco. *Information Economics and Policy*, June 2017, vol. 39(C).
Summary: Documents systematic bitcoin price differences across 11 exchanges and finds that exchanges without know-your-customer requirements are more likely to deviate from representative prices, so standard financial regulation can matter for the bitcoin market.
Skills: *Research computing:* Stata. *Data:* assembling bitcoin exchange prices and a bitcoin price index, interpolating missing prices, with FRED controls. *Econometric:* testing for unit roots (DF-GLS, KPSS) and cointegration (Johansen trace, Pesaran–Shin–Smith bounds); estimating a vector error correction model with SBIC lag selection and an LM autocorrelation test; testing Granger causality in a VAR; estimating OLS with robust standard errors and date fixed effects.

Other Publications, Reviewed Notes

6. "Cambridge Digital Mining Industry Report: Global Operations, Sentiment, and Energy Use," co-authored with Alexander Neumueller, Kamiar Mohaddes, Valentin Rousseau, and Bryan Zhang. Cambridge: University of Cambridge, CCAF, April 2025. Media coverage: The Block, CryptoSlate, The Energy Mag, BeInCrypto, Sustainability Online, Crowdfund Insider.
Summary: Surveys digital mining firms that together supplied nearly half of the Bitcoin network's computing power and estimates Bitcoin mining's annual electricity use at about 138 TWh, with about 39.8 MtCO2e of greenhouse-gas emissions. The United States is the largest mining hub, and sustainable sources supply most of the electricity, with natural gas the largest single source.
7. "Crypto-Currencies and Crypto-Assets: An Introduction," with Alexandre Olbrecht. *Eastern Economic Journal*, vol. 49, no. 2, pp. 201–205, April 2023.
Summary: The guest editors' introduction to a special issue on the economics of crypto-currencies and crypto-assets: a guide to the field and to how the issue's four papers contribute to it.
8. "3rd Global Cryptoasset Benchmarking Study," co-authored with A. Blandin, Y. Wu, T. Eisermann, A. Dek, S. Taylor, and Damaris Njoki. Cambridge: University of Cambridge, CCAF, September 2020. Media coverage: Cointelegraph, Asia Times, Bitcoin News, Cryptopotato, Crypto.news, Crowdfund Insider.
Summary: Tracks the global cryptoasset industry since the 2nd study of December 2018, from surveys of 280 entities in 59 countries: closer regulatory scrutiny after the initial coin offering bubble, new infrastructure for institutional investors, and financial engineering in mining.
9. "2nd Global Cryptoasset Benchmarking Study," co-authored with Michel Rauchs, Apolline Blandin, Kristina Klein, Martino Recanatini, and Bryan Zheng Zhang. Cambridge: University of Cambridge, CCAF, December 2018. Media coverage: Bloomberg, Cointelegraph, FXStreet, Forbes Italia, CoinGape, Crowdfund Insider.
Summary: Benchmarks the cryptoasset industry after aggregate market capitalization rose from $30 billion to over $800 billion and fell back to about $200 billion, from surveys of 127 service providers and 57 mining entities.
10. "Distributed Ledger Technology Systems: A Conceptual Framework," with Michel Rauchs, Andrew Glidden, Brian Gordon, Martino Recanatini, François Rostand, Kathryn Vagneur, and Bryan Zheng Zhang. Cambridge: University of Cambridge, CCAF, August 2018. Media coverage: The Register, Crowdfund Insider.
Summary: Proposes a formal definition of distributed ledger technology systems and the key characteristics that distinguish them from other systems, to give policymakers and developers a shared vocabulary.
Skills: *Conceptual:* applying a systems perspective from general systems theory to six case studies.
11. "The Potential Impact of Decentralized Virtual Currency on Monetary Policy." Federal Reserve Bank of Dallas, Globalization and Monetary Policy Institute, *2016 Annual Report*, April 2017, pp. 20–25.
Summary: Explains why decentralized virtual currencies such as bitcoin interest central bankers: they could change how monetary policy is administered worldwide by letting users circumvent capital controls and managed exchange rates.
Skills: *Conceptual:* charting how the virtual-currency terms relate. *Data:* building bitcoin-implied exchange rates (Pieters 2016) from bitcoin prices and official rates; using market capitalizations.
12. "Blockchain Technology Disrupting Traditional Records Systems," with Christoffer Koch. Federal Reserve Bank of Dallas, *Financial Insights*, vol. 6(2), Second Quarter 2017, pp. 1–3.
Summary: Explains how a blockchain works as a distributed register of all transactions that bypasses centralized channels such as banks, and why it has become a disruptive force in finance and elsewhere.

Skills: *Conceptual:* explaining digital signatures, block creation by solving a cryptographic puzzle, the 51% attack by a cartel of dishonest nodes, and commercial ledgers such as R3's Corda and Ripple's interledger protocol.

Book Chapters

13. "Digital Currencies and Central Banks," Chapter 6, *The Palgrave Handbook of Technological Finance*, September 2021.
Summary: Distinguishes digitized currency, digital currency, and distributed ledger technology, sets out a taxonomy of six digital-currency types, and examines the opportunities and challenges they pose for central banks, from privacy to competition between monies and monetary policy.
Skills: *Conceptual:* building a taxonomy of six digital-currency types; writing case studies of Sweden's move toward cashless payment and Kenya's M-Pesa. *Data:* using Riksbank statistics and payment surveys, Central Bank of Kenya data, and bitcoin prices.
14. "Private and Hybrid Blockchains and Applications," with Sean Stein Smith, *The Emerald Handbook of Blockchain for Business*, 2021, Chapter 6, pp. 83–98.
Summary: Defines private and hybrid blockchains against public ones with a general framework and illustrates the differences through several applications.
Skills: *Conceptual:* adapting the layer, component, and process framework for distributed ledgers from Rauchs et al. (2018); comparing applications: stablecoins against CBDCs, payment-system projects by monetary authorities, and transportation tracking.

Unpublished Papers or Work in Progress

Cryptoasset Markets and/or Blockchain-enabled Decentralization

1. "Bitcoin Reveals Exchange Rate Manipulation and Detects Capital Controls." Resting. Media coverage: The Telegraph, Bitcoin News, Finance Magnates, Breaker Mag.
Summary: Shows that bitcoin prices can approximate unofficial exchange rates and, once adjusted for bitcoin-market-specific elements, detect both the existence and the size of capital controls and exchange-rate manipulation, including transitory interventions that low-frequency data miss.
Skills: *Research computing:* Stata. *Data:* constructing bitcoin exchange rates from daily transaction-weighted prices and official rates, for 17 currencies on four exchanges; benchmarking against KAOPEN capital controls and IMF regime classifications. *Econometric:* removing trends with a Lowess smoother; testing for unit roots (ADF, KPSS) with SBIC lag selection and LM autocorrelation tests; testing cointegration (Johansen trace, Pesaran–Shin–Smith bounds) on vector and conditional error-correction models; testing Toda–Yamamoto Granger causality in a VAR.
2. "Are Cryptocurrency Markets Price Efficient? Depends on the Data Source." Resting.
Summary: Tests bitcoin price efficiency on the five largest exchanges and three price-aggregation sites and finds that the answer depends on the source: some series reject efficiency and others do not, and the disagreement persists as liquidity improves, which limits how far any empirical result generalizes.
Skills: *Theoretical economics:* applying Fama's (1970) definition of an efficient market. *Mathematical:* decomposing an aggregated price in fragmented markets. *Research computing:* Stata. *Data:* comparing price indexes with prices from the largest exchanges; selecting exchanges by trade volumes; computing log returns. *Econometric:* testing efficiency with Jarque–Bera, Box–Pierce Q, runs, and variance-ratio tests.

Quantitative Investment Research

3. "Robust Search: Exploiting Tolerance To Optimize Trading Strategies." Working paper, in progress.
Summary: Presents a parameter sweep for trading strategies that searches only among robust settings: settings that keep passing when every parameter moves at once within a declared tolerance. On a synthetic surface with two robust regions, its first two phases certify both in every run; at six parameters, none of twelve alternatives given the same budget does so in any run.
Skills: *Computational:* designing a lattice spaced by the tolerance, a local search that certifies points, and a capped search cascade; building a synthetic test surface with two robust regions, a decoy, and a spike; benchmarking against twelve alternatives, including a scrambled Sobol scatter and four global optimizers, at an equal evaluation budget.
4. "Manufactured Premia: How Universe Construction Creates the Crypto Cross-Section," co-authored with Chelsea Canon. Working paper, in progress.
Summary: Rebuilds four published crypto factor families on a point-in-time universe of the 100 largest tokens each month, booking a 100% loss when a token dies, and on a twin built the published way from end-of-sample

survivors. The size and volatility premia on the twin vanish on the point-in-time universe; momentum survives, but not a 50-basis-point trading cost.

International Macroeconomics and Trade

5. "The Effect of Income Inequality on Import Prices." Resting. An earlier version was titled "How Does Income Inequality Affect Imports?" (2017).
   Summary: Proposes a monopolistic-competition trade model with income-based price discrimination and pricing to market that replicates three documented effects of a destination's income inequality on its imports: on unit values, on trade value, and on mark-ups. Simulated, it captures half the observed elasticity of trade value to inequality and overstates the unit-value response.
   Skills: *Theoretical economics:* building a Melitz (2003) heterogeneous-firm model of trade. *Mathematical:* modeling Pareto productivity draws. *Computational:* simulating over a grid of resale constraints; solving the model by nonlinear least squares. *Research computing:* MATLAB. *Data:* assembling UN COMTRADE HS6 data on U.S. exports (2011), World Development Indicators, U.S. Census income shares, CEPII GeoDist, and the Rauch (1999) differentiation index with WITS concordances. *Econometric:* estimating trade barriers by Eaton–Kortum gravity least squares; estimating log-linear regressions with HS fixed effects and errors clustered by destination.
6. "The Composition of the Allocation Puzzle through Time." Resting.
   Summary: Decomposes capital outflows from developing countries into private savings, public debt, and reserve flows from the start of a growth episode: private outflows lead growth, public savings and reserves lag, and the allocation puzzle disappears after twenty years. None of the proposed solutions matches these facts.
   Skills: *Research computing:* Stata. *Data:* constructing capital flows from Lane–Milesi-Ferretti net foreign assets as in Gourinchas and Jeanne (2013), PPP-adjusted as in Hsieh and Klenow (2007), with Penn World Table, IMF International Financial Statistics, and World Development Indicators data; constructing labor-augmenting TFP with Caselli–Feyrer (2007) initial capital; identifying growth accelerations as in Hausmann et al. (2005). *Econometric:* estimating OLS and annual cross-sectional regressions.
7. "Re-evaluating Stylized Business Cycle Facts for Developed and Developing Countries." Resting.
   Summary: Argues that the business-cycle length most filters impose, set by Burns and Mitchell from U.S. data for 1885 to 1931, is outdated, and that using the wrong length misidentifies business-cycle patterns and distorts the imputed volatilities and correlations.
   Skills: *Computational:* dating turning points with a simplified Bry–Boschan procedure after Harding and Pagan (2002); smoothing with a 5-point Spencer curve; band-pass filtering with country-specific cycle lengths. *Research computing:* MATLAB. *Data:* benchmarking against NBER business-cycle dates; using OECD quarterly real GDP.

Economics Pedagogy and Teaching Practices

8. "Seeking Teachers: Changes in US Academic Economics Department Jobs," co-authored with Christopher J. Roark. Resting.
   Summary: Documents the growth of full-time non-tenure-track positions in U.S. academic economics departments from 2005 to 2018, to about 10% of academic economists, and a 3% annual fall in tenure-track job postings from 2014 to 2019 while postdoc postings doubled. The results question Ph.D. programs that assume a tenure-track outcome.
   Skills: *Computational:* sorting job postings by location and tenure status with keyword rules. *Research computing:* Stata. *Data:* analyzing AEA Universal Academic Questionnaire surveys, 2005 to 2018 (about 298 respondents a year), and the full text and employer-provided characteristics of AEA Job Openings for Economists postings, August 2014 to July 2019.
9. "A Guide to the U.S. Academic Economics Job Market for Teaching-Focused Jobs," co-authored with Melanie Fox, Ben Harrell, Nakul Kumar, Daniel J. Lee, & Allison Luedtke. Resting.
   Summary: A guide to teaching-focused academic jobs in economics, which differ from research jobs in several important ways: it describes the main types and how to prepare for and get them, for Ph.D. students, early-career economists, and their mentors.

Research Honors, Awards, and Recognition

2016	Awarded Trinity University Junior Faculty Sabbatical Leave ($50,000)
2014	Invited and funded attendee, Heller-Hurwicz Economics Institute, "Developing the Next Generation of Economic Models of Climate Change"
2012–15	R.R. Witt Faculty Fellowship, Trinity University

2010, 11	Travel Grant, Department of Economics, University of Minnesota
2009	Second Place, Hardy Third Year Paper Competition, for "Learning about Growth," joint with Andy Glover, Department of Economics, University of Minnesota
2009	Graduate Research Partnership Program Fellowship, University of Minnesota
2006	Graduate School Fellowship, University of Minnesota

TEACHING AND ADVISING

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(2007 to 2026) Instructor of Record Teaching Experience

Taught courses from non-calculus principles to calculus- and programming-based intermediate theory, a master's course, and a Ph.D. course, at four institutions, for classes of 8 to 400 students, in standard and accelerated formats, in person and online. Designed all homework and exams for every course. Graded all work at Trinity; at the University of Minnesota and the University of Chicago, graded some and oversaw the grading of the rest. Hired and managed teams of up to nine teaching assistants per course at a time. Intentionally selected assessment method and frequency as appropriate to the course. Redesigned course assessment and delivery for COVID-era teaching.

2026 **Collegio Internazionale Ca' Foscari, Venice**

Instructor, DoCRA Colloquium

1. Ph.D. course: Risk and Benefits of Digitized Money and Assets
   Topics: CBDC and bank stability: disintermediation, runs and flight to safety, deposit insurance, market power (deposit markdowns, the CBDC-rate sweet spot, competition regimes), feature-based demand (CES money-in-utility), payment or savings design, and reserve requirements; CBDC macro-risk consequences: monetary pass-through, the CBDC trilemma, GDP gains from bond-backed issuance, welfare and the optimal rate against cash, privacy across four regimes, and narrow central banking; stablecoins and financial risk: private money behaving as public money, peg failure, reserve demand, basket backing, and regulation (deposit insurance, capital, narrow banking); open-economy currency competition: uncovered interest parity, the impossible trinity, and first-mover spillovers; crypto monetary theory in three frameworks, positive and normative: proof of work against stake, fixed against rule-based supply, and optimal DeFi monetary and fee policy
   Skills taught: *Theoretical economics:* an OLG model of a monopolistic bank; Cournot and monopsonistic banking models of deposit-rate markdowns; Euler equations and stochastic discount factors; uncovered interest parity and open-economy algebra; no-arbitrage pricing of a pegged asset; equilibrium selection with global games; welfare analysis; time-consistency analysis, Ramsey against Markov-perfect; evaluating policy under the Lucas critique. *Mathematical:* first-order conditions; comparative statics; log-linearizing uncovered interest parity. *Computational:* calibration; re-calibrating a parameter with new data.

2018–2024 **The University of Chicago, Chicago, Illinois**

Visiting Instructor, Harris School of Public Policy (2024)

1. Macroeconomics (a macroeconomics policy course)
   Topics: method: macro against micro and partial against general equilibrium, structural against parametric models, the Lucas critique, and time-series data; policy: growth and stabilization goals, trade-offs, financing, and financial programming; measurement: GDP (value added, home and informal production, Kuznets's caution), nominal and real GDP, PPP, GNI, GNDI, and the balance of payments; growth: Malthusian to modern growth, the rule of 70, miracles, disasters, and convergence, growth accounting and TFP, and the Solow–Swan model, its steady state, transition, and consumption–investment trade-off; labor: discrimination, job fit, and TFP, unemployment measures (JOLTS, U1–U6), the Beveridge curve, and labor-market equilibrium; general equilibrium: open- and closed-economy circular flows, and import bans; fiscal policy: loanable funds, crowding out, deficits, debt sustainability, foreign-currency debt, and Ricardian equivalence; international money: exchange-rate regimes and indexes, interest arbitrage, relative PPP, the trilemma, reserves and sterilization, foreign-held Treasuries, fiscal dominance, and spillovers; monetary policy: mandates and targeting, interest-rate policy, M2, neutrality, the quantity theory, money-market equilibrium, and real rates; inflation: price indexes (deflator, CPI, PCE, PPI, trade prices; Laspeyres, Paasche, Fisher), AD–AS inflation, the Phillips curve, rational expectations, and TFP-driven cycles
   Skills taught: *Theoretical economics:* identifying the macroeconomic accounts and how they are

calculated; distinguishing agents, markets, and their interactions in a model; drawing and shifting Solow–Swan, circular-flow, present–future consumption, and leisure–consumption diagrams; applying the Solow–Swan model to named policy projects; evaluating theoretical forecasts ex post; defending true, false, or uncertain claims. *Mathematical:* choosing among logs, levels, and log-linear trends; computing a break-even oil price from IMF data. *Data:* computing total factor productivity from the Penn World Table; drawing labeled, captioned multi-series time-series graphs; interpreting time-series graphs; retrieving macroeconomic data from the World Bank, IMF, GASTAT, and UN Comtrade; classifying trade by HS commodity code; downloading, cleaning, and compiling data; cross-checking a series against several sources; choosing benchmark countries as counterfactuals; combining data and theory for policy analysis. *Econometric:* telling stationary from non-stationary and deterministic from stochastic trends, and spurious correlation from omitted variables.

Lecturer/Assistant Instructional Professor, Economics Department (2018–2024): 5 courses/year, 10–162 students per section, 1 TA per 50 students

Standard courses (9–10 week courses, offered either entirely in person or entirely online)

1. Principles of Macroeconomics (Econ 10200; no calculus or other prerequisites)
   Topics: measurement: real and nominal GDP (final and intermediate goods, per capita), its three approaches and limits, the circular flow, and stocks and flows; principles: scarcity, positive and normative analysis, the PPF, indifference curves, interdependence, and market failure; growth: the Solow model (diminishing returns, saving, depreciation, the steady state, subsistence), convergence, growth accounting and the Solow residual, TFP and discrimination, the rule of 70, and Malthusian, Kremerian, and modern growth; AD–AS: why demand slopes down, business-cycle dating, the output gap, and short- and long-run equilibrium; inflation: the deflator, the CPI and its biases, core inflation, PCE, PPI, deflation, demand and supply causes, the Phillips curve, and the sacrifice ratio; trade: rising global trade, comparative advantage, protection and Smoot–Hawley, and net exports and net capital outflow; labor: participation, unemployment types and the natural rate, unemployment insurance, minimum wages, unions, and Okun's law; schools of thought: classical to real business cycles, stagflation, sticky prices, and animal spirits; fiscal policy: stabilizers, the MPC and multipliers, the 2009 and 2020 stimulus packages, Ricardian equivalence, supply-side policy and the Laffer curve, deficits and debt, and crowding out; finance: loanable funds, interest rates, risk, maturity, bonds, and the yield curve; money: its functions, M1 and M2, fractional reserves, seigniorage, neutrality, the Fisher effect, the zero lower bound, the quantity theory, and Fed tools
   Skills taught: *Theoretical economics:* building graphical general-equilibrium intuition; decomposing general-equilibrium forces; relating the components of the macroeconomy; evaluating economic policy impacts; analyzing a shock with the Solow diagram; drawing AD–AS, circular-flow, PPF, and indifference-curve diagrams. *Mathematical:* applying algebra; calculating doubling times by the rule of 70 and convergence times with logs; deflating nominal values with price indices. *Data:* visualizing and interpreting data; reading data on log axes; predicting a series before looking at the data; retrieving FRED series and building multi-series FRED graphs; reading BLS CPI releases; using World Bank and Gapminder data. *Writing and communication:* analyzing news events and opinion pieces; critiquing a published commentary with data; defending true, false, or ambiguous claims in writing.
2. International Monetary Systems (Econ 17110; prerequisite: Principles of Macro; writing-intensive)
   Topics: exchange rates: appreciation and depreciation, volatility, bilateral, multilateral, and real rates, the J-curve, and overshooting; regimes: fixed, flexible, and dirty floats, 1873 to 1971 (the gold standard, Bretton Woods, the Nixon shock), misaligned currencies and pegs, capital flight and speculative attacks, dollarization, reserves, currency boards, sterilization, optimal currency areas, and currency unions (EMS, EMU); markets: spot, forwards, swaps, futures, options, and NDFs, arbitrage, interest parity, and the risk premium; money: the law of one price and PPP, monetary policy and exchange rates, crowding out, fiscal dominance, neutrality, and the Fisher equation; open-economy macro: exchange-rate policy in AD–AS, liability dollarization, twin deficits and crises, sovereign debt, GNI, GNDI, and the current account, the international investment position, the savings glut, and intertemporal choice; the system: trilemmas, financial globalization, global banking, the IMF, the World Bank, SDRs, and CBDCs, stablecoins, and cryptocurrencies
   Skills taught: *Theoretical economics:* building the economic representation of a scenario; solving the representation under different scenarios; extrapolating the implications for economic agents; drawing money-market and foreign-exchange graphs; evaluating practitioners' statements with theory. *Mathematical:* deriving first-order Taylor approximations of the parity conditions; computing year-on-year percent changes, averages, and medians. *Research computing:* graphing downloaded data in a spreadsheet or statistical package. *Data:* assembling FRED exchange-rate and interest-rate series; comparing data with theory for at least three variables; sketching the time-series implications of a model. *Econome-*

ric: testing interest-rate parity against data. *Writing and communication:* writing policy reports within a 750-word cap; finding and summarizing a policy source; reading financial-press articles; citing central-bank and academic literature; drafting sections for feedback.

3. The Elements of Economic Analysis I (Econ 20000; Intermediate Micro: calculus-based consumer theory)
   Topics: foundations: scarcity, price-taking, and purposive behavior; budget sets: endowments, price changes, rationing, taxes, subsidies, the numeraire, and opportunity cost; preferences: the axioms, indifference curves and the MRS, special cases (substitutes, complements, neutral goods, bads, bliss points, non-convex and lexicographic preferences), and ordinal and cardinal utility; demand: utility maximization and duality, Marshallian and Hicksian demand, indirect utility and expenditure functions, Roy's identity, Shephard's lemma, the multiplier as the marginal utility of income, and the diamond–water paradox; comparative statics: the law of demand, Slutsky income and substitution effects, Giffen, normal, and inferior goods, gross and net substitutes, Hicksian symmetry, revealed preference, and Engel curves; welfare: compensating and equivalent variation, consumer surplus, deadweight loss, and the endowment effect; time: labor–leisure choice, home production, and pricing non-market activity (crime, marriage, fertility, voting); intertemporal choice: discounting, independent wants, storage, credit markets, the Euler equation, and permanent income; uncertainty: expected utility, risk aversion, insurance, and moral hazard (the Peltzman effect); general equilibrium: the Robinson Crusoe endowment economy, Walras' law, the Edgeworth box, gains from trade, Pareto optimality, the contract curve, and the core
   Skills taught: *Theoretical economics:* building a consumer-choice problem from a scenario; translating the problem across analysis settings; solving the problem under different specifications; measuring the welfare effects of taxes and subsidies; drawing indifference curves, budget lines, and the Edgeworth box; proving results graphically with Marshallian and Hicksian demand. *Mathematical:* applying multivariable calculus; solving with Lagrangians and first-order conditions, or by substitution; testing cases with Karush–Kuhn–Tucker conditions; calculating elasticities; integrating under the demand curve.
4. The Elements of Economic Analysis II (Econ 20100; Intermediate Micro: calculus-based producer theory)
   Topics: technology: production functions (Cobb–Douglas, fixed-proportions, linear, CES), their properties, marginal and average product, Edgeworth complements, output elasticity, returns to scale, technical change, isoquants, the MRTS, and the elasticity of substitution; costs: fixed, variable, and sunk costs, cost curves, the short and long run, economies of scale, cost minimization, conditional factor demands, and Shephard's lemma; the competitive firm: profit maximization, Hotelling's lemma, MR = MC, natural monopoly, supply, shutdown, and exit; markets: industry supply, free entry, intensive and extensive margins, interventions (taxes, subsidies, quotas, price controls), incidence, and surplus and deadweight loss; general equilibrium: the Robinson Crusoe production economy, the PPF, Walras' law, the planner, the representative agent and the Solow residual, social welfare functions and Arrow's theorem, and the welfare theorems; market failure: externalities, Pigouvian taxes, cap-and-trade, the Coase theorem, and public goods; market power: monopoly, the Lerner index, and price discrimination; games: Nash equilibrium, dominant and mixed strategies, the prisoner's dilemma, zero-sum games, Bertrand, Cournot, and Stackelberg competition, collusion and repeated games, and subgame perfection; information: signaling, left-digit bias, optimal income taxation, and mechanism design
   Skills taught: *Theoretical economics:* building a firm-choice or general-equilibrium problem from a scenario; translating the problem across analysis settings; solving the problem under different specifications; finding Nash equilibria, best-response functions, and mixed strategies; solving sequential games by backward induction; converting normal-form games to extensive form; performing graphical comparative statics; drawing isoquant and cost-curve graphs; evaluating a news claim with the model. *Mathematical:* solving with Lagrangian, first-order, and Karush–Kuhn–Tucker conditions; finding market-clearing prices with the quadratic formula.
5. The Elements of Economic Analysis III (Econ 20200; Intermediate Macro: DGE + beginning MATLAB programming)
   Topics: saving and assets: the two-period model, present value, permanent income, the Euler equation, credit limits, bond markets and the yield curve, general-equilibrium interest rates, equity under uncertainty, complete markets, and the CAPM, C-CAPM, and equity premium puzzle; growth: growth accounting and TFP, human capital and the Lewis turning point, Malthusian, Kremerian, and modern growth, Kaldor's facts, the Solow–Swan model and the golden rule, and the neoclassical growth model; equilibrium: the representative agent, the planner, the welfare theorems, and GDP and the Lucas critique; business cycles: consumption–labor choice under productivity shocks, trend and cycle facts, real and international real business cycles, and the cost of cycles; fiscal policy: multiplier disputes, Ricardian equivalence, and hand-to-mouth households; labor: unemployment measures, search, the Beveridge curve, Okun's law, and the Phillips curve; money: cash-in-advance and OLG models, Baumol–Tobin

demand, the quantity theory, neutrality, sticky prices, Nash and Ramsey policy equilibria, inflation targeting, Cagan hyperinflation, seigniorage, the Fisher equation, the zero lower bound and negative rates, open market operations, fiscal dominance, and unpleasant monetarist arithmetic
Skills taught: *Theoretical economics:* constructing and solving general-equilibrium models and testing them against data; comparing the social planner's problem with the decentralized market; modifying a theory to evaluate statements or scenarios; drawing Solow and phase diagrams. *Mathematical:* solving with Lagrangian methods and first-order conditions; deriving Euler equations; applying the transversality condition; approximating growth rates; discounting to present value. *Computational:* calibrating a parameter to match data. *Research computing:* computing and plotting model equilibria in MATLAB; solving models in Dynare. *Data:* computing the return to a college degree from FRED data; adjusting units to build per-capita series.

6. Economic Policy Analysis (Econ 23950; Intermediate Macro: calculus- and programming-based (MATLAB or Python), writing-intensive)
Topics: foundations: fiscal and monetary policy, theory and measurement, the Lucas critique, the welfare theorems, planner against market, static and dynamic, partial and general equilibrium, RBC facts, the history of macroeconomics, and policy objectives; taxation: lump-sum taxes in steady state and transition, efficiency, proportional taxes on income, capital, labor, and consumption, the Laffer curve, consumption-equivalent welfare, Ramsey taxation and the inverse-elasticity rule, time inconsistency, Mirrlees taxation and incentive compatibility, incidence, and compliance, framing, and enforcement; spending: Keynesian and classical multipliers and the MPC, crowding out, productive spending, Ricardian equivalence and its failure, permanent income, and hand-to-mouth households; intervention: price controls, the minimum wage, monopoly and monopsony, and Pigouvian taxes; debt: the government budget constraint, optimal and foreign-currency debt, and default; redistribution: the Edgeworth box, social welfare functions, pay-as-you-go and funded social security in OLG models, unemployment insurance, and McCall search; money: medium of exchange (Kiyotaki–Wright, barter, the double coincidence of wants), fiat and commodity money, the optimal supply, store of value (cash-in-advance, money-in-utility, Baumol–Tobin, the Friedman rule), the Fisher equation, neutrality and superneutrality, unit of account, and the quantity theory; nonneutrality: sticky prices and wages, the Lucas island model, the Phillips curve, adaptive and rational expectations, and stagflation, Volcker, and the sacrifice ratio; seigniorage: fiscal dominance, the seigniorage Laffer curve, unpleasant arithmetic, and Cagan hyperinflation; international: pre-Fed US monetary history, the gold standard, Bretton Woods, the Nixon shock, exchange-rate regimes, interest parity, the trilemma, and Bitcoin-implied exchange rates; central banking: commitment and the dual mandate, inflation targeting, communication, reserve requirements and the money multiplier, open market operations, Treasuries and inversion, the federal funds rate, interest on reserves, the reverse repo, the discount window, the zero lower bound, and the Taylor rule; cases: universal basic income, wealth taxes, the mortgage-interest deduction, the CARES Act, MMT, competing currencies, cryptocurrencies, and CBDCs
Skills taught: *Theoretical economics:* computing elasticity comparative statics of the steady state; comparing policies by consumption equivalent variation; proposing a model of a policy proposal; modifying models to evaluate policy under different environments or assumptions; applying general-equilibrium models to monetary policy. *Mathematical:* writing the social planner's Lagrangian; deriving Euler equations; using Bellman equations and dynamic programming; using integration by parts for expectations over a wage distribution; summing present discounted values as geometric series. *Computational:* simulating tax policy over 200 periods in MATLAB; checking robustness by looping over parameters; synthesizing simulated predictions with data. *Research computing:* simulating in MATLAB and Dynare; writing loops in MATLAB or Python. *Data:* adjusting data for inflation and population; building scatter plots and graphs from downloaded data. *Econometric:* testing the Phillips curve on FRED data; telling correlation from causation. *Writing and communication:* finding and summarizing primary-source policy documents; working from a Federal Reserve research article; analyzing news events and opinion pieces with economic methods; writing three policy briefs; writing a policy paper with simulations in MATLAB, R, or Python, a code appendix, and graphs from data.

Accelerated courses (5- to 6-week courses)

7. The Elements of Economic Analysis II (Econ 20100; calculus-based producer theory, intermediate micro)
Same topics and tools as the standard course.
8. Economic Policy Analysis (Econ 23950; calculus- and programming-based (MATLAB or Python), writing-intensive)
Same topics and tools as the standard course.

Co-taught courses

9. MA Thesis Development Workshop (macroeconomics and finance specialization)
   Topics: the question: literature review and pruning, pitches that are interesting, refined, and feasible, progress reports and cutting losses, the value of a yes or no answer, and its anatomy (decision makers, constraints, equilibrium, qualitative or quantitative, internal and external validity); the design: a critical review of the closest papers, model, data, and identification (exclusion restrictions, instruments); the thesis: proposal.
10. Pathways in Economics, Non-calculus (Econ 12410; High School Summer Program)
    Topics (macroeconomics block)**:** perspectives: economics as a science, inequality, the Phillips curve and the macroeconomic revolution, the Lucas critique and structural models, and behavioral macroeconomics; measurement: national accounts, the circular flow, the deflator, CPI, and PCE (and Social Security indexation), saving, investment, and net exports, GDP against GNP, and unemployment, participation, and wages; growth: Kaldor's facts, the rule of 70, the Solow model (steady state, transition, the golden rule, technology and balanced growth), endogenous growth and human capital, and convergence; choice: consumption–leisure and labor-market equilibrium, income and substitution effects, the two-period model with borrowers and lenders, and loanable funds; cycles: NBER dating, stylized facts, TFP shocks, consumption smoothing and permanent income, investment volatility, labor over the cycle, and AD–AS; fiscal policy: spending and tax facts, lump-sum and proportional taxes, crowding out, the budget constraint and Ricardian equivalence, the Keynesian cross against the supply-side view, spending against transfers, automatic stabilizers, austerity (the US and Greece, 2009), and optimal procyclical taxes and countercyclical spending; money: real and nominal variables, its functions, Baumol–Tobin, supply, demand, and neutrality, M1, M2, and MZM, the multiplier, the quantity equation, and seigniorage; central banking: goals, open market operations and the federal funds rate, the Friedman and Taylor rules, the zero lower bound, interest on reserves, the reverse repo, quantitative easing, and forward guidance; open economy: interest parity, the impossible trinity, and tariffs
    Skills taught: *Theoretical economics:* setting up a household optimization problem and deriving its first-order condition; analyzing a Laffer curve; drawing indifference-curve, budget-constraint, labor-market, AD–AS, and money-market diagrams; plotting Solow transition paths. *Mathematical:* calculating compound growth and doubling times; summing the multiplier as a geometric series; computing GDP by the value-added approach; differentiating implicitly. *Data:* graphing FRED data per capita and as shares of GDP; estimating a policy saving from two inflation measures; relating the federal funds rate to unemployment and inflation; exploring a data visualizer. *Writing and communication:* fact-checking an op-ed; critiquing news pieces; summarizing Federal Reserve publications and proposing research questions; researching historical episodes; defending true, false, or uncertain answers.
11. Pathways in Economics, Calculus-based (Econ 12411; High School Summer Program)
    Same topics and tools as Econ 12410, with single-variable calculus. Adds optimal distortionary taxation, including a zero capital tax and the time-inconsistency problem, and a cash-in-advance model of money.
12. A Survey of Chicago Economics (Econ 12412; Specialized Summer Survey Program)
    Same topics and tools as Econ 12410.

Financial Mathematics Program: a summer overload, outside the Economics Department appointment

13. Cryptoasset Markets (master's course, FINM 31000, a 5- to 6-week accelerated course)
    Topics: foundations: centralized and decentralized currencies, blockchain and DLT layers (token, network, protocol, data), consensus (proof of work, mining, the 51% attack, proof of stake), MEV, and decentralization in practice; history, 2009 to 2022: the Blocksize War, The DAO, crypto winter, the DeFi and NFT booms, Ethereum and smart contracts, forks (Ethereum Classic), and bridges and wrapped tokens; regulation: who regulates and regulatory arbitrage, ICOs, IEOs, and STOs, taxation and trading in the US and Germany, and tokens as securities; tokens: altcoins, dApps and ERC-20, transaction, utility, governance, and security tokens, and NFTs; valuation: quantity-theory bounds on bitcoin, token against project value, NVT, MVRV, and Metcalfe's law, and layer-based valuation; returns: calendar effects, distributions and volatility against stocks, factor exposures (currencies, gold, the US macroeconomy, CAPM to Fama–French, geography, industry), fundamentals (energy and computing costs, regulation, network factors), momentum and its sources, attention, sentiment, and a trading-strategy project; market structure: centralized and decentralized exchanges and their hacks (Mt. Gox, Bitfinex, Poly Network), retail against institutional trading, derivatives above spot volume, on-chain, off-chain, and exogenous data, and data reliability (fake volume, wash trading, the Mt. Gox bots, Tether); trading: pump-and-dumps, spatial and triangular arbitrage, perpetual futures and funding-rate arbitrage, and staking; DeFi: oracles, DAOs, total value locked, Uniswap and constant-product AMMs, impermanent loss, yield farming, DAI and liquidations, and reserve-backed and algorithmic stablecoins (Terra–Luna, multiple equilibria)
    Skills taught: *Theoretical economics:* building Markowitz portfolios and maximizing Sharpe ratios. *Con-*

ceptual: evaluating a token's role and market niche; classifying tokens as transaction, utility, governance, or security tokens; analyzing how tax law shapes trading. *Computational:* forecasting prices with uncertainty bounds; calculating arbitrage profits net of fees and transfer times; designing, backtesting, and coding a trading strategy on API data without look-ahead. *Research computing:* pulling exchange data through APIs in Python; submitting code and data with each project and citing data sources. *Data:* evaluating the promise and limits of crypto data; looking up sector and usage data. *Econometric:* regressing returns on same-month and lagged fundamentals and on search attention; testing for momentum; analyzing time series; building correlation and covariance heat maps; signing expected correlations; replicating published analyses on a new token.

2012–2018 **Trinity University, San Antonio, Texas**

Assistant Professor, Department of Economics: 6 courses/year, 8–35 students, no TAs or graders

1. Principles of Microeconomics (no calculus or other prerequisites)
2. Principles of Macroeconomics (no calculus; Micro principles prerequisite)
   Topics: AD–AS: the circular flow, market and command economies, aggregate demand, its slope and shifters, the multiplier and MPC, short- and long-run supply, sticky wages and prices, the natural rate of output, and shocks; measurement: real and nominal GDP, its expenditure components, and the CPI, inflation, and their measurement problems; growth: real GDP per capita, cross-country comparisons, and policy (institutions, human capital, saving, trade, patents); cycles and labor: the GDP gap, unemployment types and the natural rate, US recessions and NBER dating, and the case for stabilization; finance: real and nominal rates, risk and efficient markets, and loanable funds; money: its functions and forms, the gold standard, money demand and equilibrium, fractional reserves and the multiplier, policy tools (open market operations, the reserve ratio, the discount rate, interest on reserves), and the quantity theory; policy: the Phillips curve and the natural-rate hypothesis, rules against discretion, the Taylor rule, central-bank independence, fiscal stabilizers and timing problems, the Laffer curve, deficits and debt, and crowding out; trade: net capital outflow, trade balances, absolute and comparative advantage, the terms of trade, gains from trade, arguments for barriers, tariffs and quotas, nominal and real exchange rates, PPP, and exchange-rate systems
   Skills taught: *Theoretical economics:* computing consumer, producer, and total surplus; drawing supply–demand, AD–AS, circular-flow, loanable-funds, and Laffer diagrams. *Mathematical:* solving linear equations; computing growth rates, rule-of-70 doubling times, and compounding; computing index numbers and real GDP; calculating production-function growth; building T-account bank balance sheets and computing money multipliers; computing spending multipliers from the marginal propensity to consume; computing comparative advantage from opportunity-cost tables; computing risk-adjusted returns of college choices; ordering loan repayments under compound interest. *Research computing:* working with data in Excel, visualizing it and fitting trendlines. *Data:* using FRED, Gapminder, and College Scorecard data; using Department of Labor minimum-wage history, the Minneapolis Fed CPI calculator, and state tax and federal-spending tables; merging two sources into a ratio, converting units; refuting a policy claim with time-series plots.
3. International Trade (no calculus; Macro principles prerequisite; writing-intensive: 15-page research paper)
   Topics: foundations: the trade balance, autarky, free trade, protectionism, mercantilism, classical against new trade theory, and welfare as real wages; models: relative supply and demand, absolute and comparative advantage, Ricardian trade, specific factors (real wages and rents), Heckscher–Ohlin, Stolper–Samuelson, Rybczynski, migration and wage equalization, and scale economies with monopolistic competition; evidence: Ricardian and Heckscher–Ohlin tests, the Leontief paradox, intra-industry trade, and gravity and border effects; policy: tariffs and quotas under competition and monopoly, customs unions and free-trade areas, trade agreements, tariff games, subsidies (Boeing and Airbus), dumping, non-tariff barriers, and the WTO; development: import substitution and infant industries, export-led growth, and Singer–Prebisch; globalization: offshoring, outsourcing, FDI, and portfolio investment
   Skills taught: *Theoretical economics:* solving the Ricardian model; working specific-factors and Heckscher–Ohlin comparative statics; finding Nash equilibria in normal-form payoff matrices; drawing tariff graphs; testing which trade model fits a good's trade flows; testing the Singer–Prebisch hypothesis. *Mathematical:* computing the intra-industry trade ratio; normalizing a price index. *Data:* visualizing trade data with log transforms; navigating World Bank, OECD, USA TradeStats, WITS, and BLS data; classifying trade by Harmonized System code; collecting a trade data set of at least 40 points. *Econometric:* running and interpreting single and multivariate regressions in Excel. *Writing and communication:* writing a term paper with a proposal, weekly drafts, and an abstract; discussing literature from central banks and academic economists; peer-reviewing drafts in class; writing and answering exam questions;

teaching the class in an optional presentation.

4. International Monetary Systems (no calculus; Micro principles prerequisite; writing-intensive: 15 pages)
   Topics: exchange rates: appreciation, effects on profits and trade prices, the foreign-exchange market and arbitrage, foreign returns and risk, the asset approach, the risk premium, and herding; money: money, interest rates, and the exchange rate, long-run neutrality, the law of one price, PPP, the Fisher effect, and overshooting; short run: Mundell–Fleming (IS-LM-FX), the liquidity trap and zero lower bound, the trilemma, and capital controls; pegs: emerging-market depreciation and capital flight, defense by reserves, sterilization, and currency boards, crises, the reasons to peg, and liability dollarization; systems: the gold standard, Bretton Woods, and after; globalization: its gains (smoothing, investment, diversification), national accounts (GDP, GNP, GNDI, the current account), original sin and default, global banking and contagion, Basel capital rules, currency areas, the EMS, EMU, euro criteria, and the Greek crisis, and the IMF (quotas, conditionality, voting, the Troika) and the World Bank; cryptocurrencies: their history, blockchain and mining, exchanges, ICOs, forks, government digital currency, financial stability, money laundering, sanctions evasion, and monetary-policy effectiveness
   Skills taught: *Theoretical economics:* applying a model to a current news article and testing its policy prescriptions; classifying exchange-rate regimes from graphs; reconciling policymakers' statements. *Mathematical:* deriving arbitrage conditions with fees; calculating interest parity; calculating expected returns under risk; calculating currency conversions and profits under fixed and flexible rates; calculating money growth and inflation and drawing their time paths. *Data:* graphing FRED and World Bank data in Excel; retrieving daily exchange rates; explaining deviations from theory with data. *Econometric:* testing the law of one price by OLS (a paper option); running Excel regressions with trendlines and p-values. *Writing and communication:* writing a term paper with weekly drafts; citing literature from central banks, government economists, and academic economists; peer-reviewing drafts in class; writing and answering exam questions; giving an optional presentation.

2007–2012 **University of Minnesota, Minneapolis, Minnesota**

Lead Instructor, Principles of Macroeconomics (2010–2012)

1. Principles of Macroeconomics (created and maintained a uniform curriculum across 8 instructors)

Instructor (2007–2012): 3 courses/year

1. Principles of Macroeconomics: 2009–2012, Large Lecture Instructor (150–400 students, 2–9 TAs)
2. 2008: Intermediate Macroeconomics (calculus-based) (15–40 students, 0–1 TAs)

Teaching Assistant (2007–2008)

1. Principles of Microeconomics and Principles of Macroeconomics

Student-Research Advising Experience

2018–2024 The University of Chicago: advising Economics MA students (3–6 master's students/year; master's-level thesis required; placements include UCLA, NYU, and HKU Ph.D. programs); advising undergraduate honors theses (0–1 students/year)

2012–2018 Trinity University: advising undergraduate thesis or research papers, economics or economics-adjacent (0–2 students/year)

Notable Supervised Undergraduate Research Outcomes

2019 D. Gale Johnson Award for undergraduate honors thesis ("A Nation of Makers: Endogenous Innovation in Makerspaces" by Won Woo Kim)

2014 3× acceptance, Economics Scholar Program (presentation and poster sessions); 1× Southwest Social Science Association (presentation)

In-Discipline Mentoring

2019–2024 Faculty Advisor, University of Chicago Undergraduate Women in Economics RSO

2019–2020 Research in Color Foundation: undergraduate research mentor for underrepresented students in economics (1 student; research poster presentation required)

2019–2020	Presenter: The University of Chicago College Day Lecture for the Chicago Public High School "Bridge to College" program
2018–2020	Mentor: CSWEP Breakfast for Early Career Economists
2012–2018	Trinity University: advising undergraduate International Economics or General Economics majors (4–8 students/semester); general advising of undeclared-major students (first years until major declaration: 7–10 students/semester)

Teaching Honors and Awards*

University of Minnesota, Minneapolis, Minnesota

2011	Walter W. Heller Outstanding Teaching Award, College of Liberal Arts
2010	Walter W. Heller Outstanding Teaching Award, College of Liberal Arts
2008–2012	Distinguished Instructor, awarded 8 times
2008	Distinguished Teaching Assistant

*Not eligible for teaching awards at Trinity University or The University of Chicago.

ACADEMIC CONFERENCE OR RESEARCH SEMINAR PRESENTATIONS

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(self or co-author presentation of original work titled as listed, unless otherwise indicated)

Cryptoasset Markets and/or Blockchain-enabled Decentralization

"Cryptocurrencies, Stablecoins, and Central Bank Digital Currency," session panelist

- 2020: Southern Economic Association (online)

"How Have Cryptocurrency Participants Changed?"

- 2019: American Economic Association (Atlanta Meeting); Texas State University

"How Global Is the Cryptocurrency Market?"

- 2018: FRB Atlanta: Workshop on Financial Stability Implications of New Technology; Western Economic Association (Vancouver, Canada); International Trade and Finance Association (Beijing, China)

"Cryptocurrencies As a New Global Financial Asset"

- 2018: American Economic Association (Philadelphia)

"Bitcoin Reveals Exchange Rate Manipulation and Detects Capital Controls" (various working titles)

- 2019: Lafayette College
- 2018: De Nederlandsche Bank (DNB); Royal Economic Society Conference (University of Sussex, UK); Hobart and William Smith Colleges
- 2017: Federal Reserve Bank of Dallas; Midwest Macroeconomics Meeting (Fall, University of Pittsburgh); The 49th Money, Macro and Finance Conference (King's College, London); Liberal Arts Macroeconomics Workshop (Davidson College); Society for Economic Measurement (MIT)
- 2016: Eastern Economic Association (Washington D.C.); International Trade and Finance Association (Occidental College)
- 2015: Southern Economic Association (New Orleans)

"Financial Regulations and Price Inconsistencies across Bitcoin Markets"

- 2016: Western Economic Association (Portland)

International Macroeconomics and Trade

"The Effect of Income Inequality on Import Prices" (earlier titled "How Does Income Inequality Affect Imports?")

- 2017: Midwest Macroeconomics Meeting (Spring, Louisiana State University); Western Economic Association (San Diego)

Economics Pedagogy and Teaching Practices

"Authentic Assessments"

- 2025: De La Salle University – Manila

"Relevance, Belonging, and Growth in Economic Education," discussant

- 2025: American Economic Association (San Francisco) [session by AEA Committee on Economic Education]

"Assessing and Using Prior Student Knowledge in 9 Weeks," panelist

- 2024: The University of Chicago Center for Teaching and Learning

"Teaching Courses in Macroeconomics and Monetary Policy Through Active Learning," discussant

- 2023: AEA Conference on Teaching and Research in Economic Education (Portland)

"Practical Assessment Design: How to Make Assignments Work for You and Your Students"

- 2023: The University of Chicago Center for Teaching and Learning

"The Ancillaries of Undergraduate Economics Program: The Extras Can Make All the Difference," discussant

- 2023: American Economic Association (New Orleans) [session by AEA Committee on Economic Education]

"Avalanche! A Method to Facilitate Discourse in Classes of Any Size"

- 2022: Southern Economic Association (Fort Lauderdale) [Presidential Session]

"Actively Selecting Assessment Techniques: Asking the Why"

- 2022: American Economic Association (Boston)

"Inspiring Ideas in Economics Education," session chair

- 2021: TeachECONference 2021

"Exploratory Teaching Groups Colloquium, Economics Teaching Group"

- 2021: The University of Chicago Center for Teaching and Learning

"Seeking Teachers: Changes in US Academic Economics Department Jobs" (original title "Changes in US Academic Economics Departments")

- 2021: Southern Economic Association (Houston); Robert Morris University 32nd Annual Teaching Economics Conference
- 2020: Troy University Teaching Symposium in Economics (cancelled due to COVID)

"Social Media Broadens Reach: Our Responsibilities in an Open Classroom," panel chair and organizer

- 2021: AEA Conference on Teaching and Research in Economic Education

"Formative and Summative Assessment Strategies"

- 2019: The University of Chicago Center for Teaching and Learning

"Efficient Strategies for Writing Effective Exams"

- 2019: The University of Chicago Center for Teaching and Learning

Funded, Invitation-Only and/or Competitive Professional Development Workshops

2023	EnCOREage Workshop, Center for Advanced Study in the Behavioral Sciences, Stanford University
2022–2023	Associate Fellow (aka Mentoring/Sr. Fellow), Pedagogy Fellows Program, Chicago Center for Teaching and Learning (CCTL), The University of Chicago
2021–2022	Pedagogy Fellow, Pedagogy Fellows Program, CCTL, The University of Chicago
2019	Wharton Cryptogovernance Workshop, San Francisco
2017	Liberty and the War on Cash, sponsored by Liberty Fund, San Antonio
2015	CeMENT Mentoring for Junior Faculty at Liberal Arts, sponsored by CSWEP (Committee on the Status of Women in the Economics Profession), New Orleans

PUBLIC SERVICE: CLASS GUEST SPEAKER, NON-ACADEMIC PRESENTATIONS OR PANELS

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Presentations or Panel Member for a Non-Academic/Policymaker Audience

2026	NABE Chicago, panel moderator, "New Rules & Landscapes: Stablecoins, the GENIUS Act, and the Future of Payments" (October 5)
2026	EconVue, panel member, "Stablecoins: The View from Chicago" (January 7)
2025	Monetarium 2 (Washington DC)
2023	DFW Association for Business Economics (online); R3 CordaDay (New York City)
2021	Chicago Financial Forum; Wharton Crypto-Governance Group (online)
2020	Chicago Financial Exchange (online)
2019	DEconomy (Seoul, South Korea); Chicago University Blockchain Symposium (Northwestern University)
2018	Blockchain & Cryptocurrency Education Sessions at Geekdom (San Antonio); University of Cambridge, Centre for Alternative Finance launch of Global Cryptoasset Reports (hosted by CME Group, Chicago); National Association for Business Economics (NABE), Dallas Chapter; San Antonio Texas A&M Club; FinTech DC (IMF, Washington DC)
2017	San Antonio Business and Economics Society (SABES) luncheon speaker
2015	Faculty research presentation for Delta Sigma Pi Fraternity, Trinity University

Invited Guest Speaker in a University Class

2021	The Ohio State University, Principles of Macroeconomics
2020	The University of Chicago, LAWS 42505: Blockchain and Cryptocurrencies; The University of Chicago, Econ 13000: Money and Banking; Pitzer College, Economics Senior Seminar
2019	University of Illinois at Urbana-Champaign, Finance 490: Fintech; Pitzer College, Economics Senior Seminar; The University of Chicago, Econ 13000: Money and Banking; The University of Chicago, Econ 23040; The University of Chicago, Cryptocurrencies; The University of Chicago, MPCS 56600: Intro to Blockchain (MPCS = Masters Program in Computer Science)
2018	The University of Chicago, Research Experience for Undergraduates (Economics Department)

PUBLIC SERVICE: MEDIA

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Media Interview: Radio, Podcast, or Television

1. BBC: Outside Source, "The current crash in the crypto markets" (live television interview, June 13, 2022)
2. Canadian Broadcasting Corporation (CBC) Tai Asks Why, "What will money look like in the future?" (April 27, 2022)
3. BBC Business Daily Podcast, "Why does Bitcoin consume so much energy?" (February 26, 2021)
4. Made in Tech, "Vibrant cryptocurrency growth has excited many. Can regulation spoil it?" (February 12, 2021)
5. British Broadcasting Corporation (BBC): The Digital Human, "Oops!" (February 24, 2020)
6. WBEZ Chicago – Worldview: "What Is Blockchain Technology and Why Should We Care?" (July 30, 2018)
7. Scholars' Circle Radio Show, "Cryptocurrency, Insights from Scholars" (April 15, 2018)
8. Texas Public Radio (TPR) The Source, "Cash, Credit, Cryptocurrency: How Do You Choose to Pay?" (panel member on a call-in show, January 2, 2018)
9. National Public Radio (NPR) Marketplace: "Why pay a ransom in bitcoin?" (May 15, 2017)
10. NPR Marketplace Tech: "The rise of cryptocurrencies" (June 22, 2017)

Media Interview: Written or Quoted

1. Bloomberg Government: "Texas seeks 'Crypto Capital' status among state Bitcoin Reserves" (May 22, 2025)

2. BBC News: "Is the US trying to kill crypto?" (June 15, 2023)
3. Global Finance: "Central Bank Digital Currencies at a Crossroads" (February 6, 2023)
4. Chicago Tribune: "FTX collapse shakes up Chicago crypto market" (December 12, 2022)
5. CoinDesk: "Crypto Goes to College" (September 29, 2022)
6. BBC News: "Will the US crack down on cryptocurrency?" (March 9, 2022)
7. Cointelegraph Magazine: "Crypto as a 'public good' in the 22nd Century" (October 22, 2021)
8. Cointelegraph Magazine: "Is the cryptocurrency epicenter moving away from East Asia?" (August 10, 2021)
9. Cointelegraph: "Terrorists still raise money through crypto, but the impact is limited" (April 9, 2021)
10. BBC Science: "How Bitcoin's vast energy use could burst its bubble" (February 27, 2021)
11. Cointelegraph: "Happy Birthday Dear Bitcoin: Crypto's first white paper turns 12" (October 2020)
12. Nikkei: "Digital currency collaborative research, interviews with American experts" (January 22, 2020)
13. The Block: "Mad Crypto: Hypothesizing bitcoin in a downcycle" (August 7, 2019)
14. Breaker Mag: "Economists Are Missing a Trick With Crypto" (April 15, 2019)
15. The Street: "This Is Why Fidelity Is Diving Into Bitcoin Even Though It Has $6.2 Trillion in Assets" (August 10, 2017)
16. Experian: "What is Bitcoin and should you buy it?" (September 12, 2017)

Opinion Pieces

1. CSWEP News: "The Job Market for Non-Tenure Track Academic Economists" (Issue II, 2022), co-authored with Dr. Chris Roark
2. CoinDesk: "What Is a Cryptocurrency? We Need Clearer Definitions" (December 31, 2019)

SERVICE TO THE DISCIPLINE

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Editor, Selection Committee Member, or Session Organizer for Conferences and Journals

2025	Conference Selection Committee member: The Crypto Asset Lab Conference – Investments, Economics, and Regulation for Bitcoin, Crypto-Assets, and CBDCs (hosted by the University of Milano-Bicocca, joint with DG Joint Research Centre, European Commission)
2024	Conference Selection Committee member: The Crypto Asset Lab Conference – Investments, Economics, and Regulation for Bitcoin and Crypto-Assets (hosted by the University of Milano-Bicocca, joint with DG Joint Research Centre, European Commission)
2023	Conference Selection Committee member: 3 × AEA 2024 CSWEP sessions; Conference Selection Committee member: Tokenomics: International Conference on Blockchain Economics, Security, and Protocols (hosted by Columbia University)
2022	Co-Editor: special edition of the *Eastern Economic Journal*, "The Economics of Cryptoassets"
2021	Conference Selection Committee member: The Crypto Asset Lab Conference – Investments, Economics, and Regulation for Bitcoin and Crypto-Assets (hosted by the University of Milano-Bicocca, joint with DG Joint Research Centre, European Commission)
2019	Conference Selection Committee member: Tokenomics: International Conference on Blockchain Economics, Security, and Protocols (hosted by École Normale Supérieure); Conference Selection Committee member: International Trade and Finance Association Annual Conference (hosted by Università di Pisa); Panel Organizer: DEconomy Panel on Central Bank Digital Currencies
2018	Session Organizer: International Banking, Economics and Finance Association cryptocurrency sessions (2) at the Western Economic Association meeting (Vancouver, Canada)

Referee/Reviewer Service

British Journal of Management • Computational Economics • Czech Science Foundation • Eastern Economic Journal • Economics Bulletin • Economic Theory • European Accounting Review • European Journal of Finance • Frontiers in Blockchain Research • Global Finance Journal • Information Economics and Policy • International Review of Economics Education • Journal of Economic Insight • Journal of International Economics • Journal of International Money and Finance • Sage Open • Swiss National Science Foundation • Quarterly Review of Eco-

nomics and Finance

Textbook Reviewer

- Sachs, Jeffrey and Gordon McCord, 2009: *Principles of Economics and Global Sustainable Development*, Addison-Wesley: Boston (prospectus review)
- Cowen, Tyler and Alex Tabarrok, 2009: *Modern Principles of Economics*, Worth Publishers: New York
- Williamson, Stephen, 2009: *Macroeconomics*, 3rd edition, Addison-Wesley: Boston (pre-revision review)

SERVICE TO THE UNIVERSITY

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The University of Chicago

2023–2024 Committee Chair: Econ MA Thesis (3 papers); Committee Member: Econ MA Thesis (1 paper); Faculty Advisor: Blockchain Chicago RSO; Faculty Advisor: Undergraduate Women in Economics RSO; Letters of Recommendation (13 students, most with multiple letters); Mentor: Professional Development Lunch for Female PhD Students, Post Docs, Senior Research Associates and Faculty; Organizer: oversaw undergraduate and MA department TA and grader hiring; Organizer: Fall Macro MA Thesis Workshop; Reviewer: MAPSS-Econ applications (79 applications); Preceptor-Advisor: Econ MA students (6 advisees); MA Thesis Advisor (4 advisees)

2022–2023 Associate Fellow: Chicago Center for Teaching and Learning; Committee Member: Ad-Hoc Committee on Exam Accommodation for Students with Disabilities; Committee Member: Econ MA Thesis (3 papers); Committee Member: Public Policy MA Thesis (1 paper); Faculty Advisor: Blockchain Chicago RSO; Faculty Advisor: Undergraduate Women in Economics RSO; Letters of Recommendation (22 students, most with multiple letters); Mentor: Professional Development Lunch for Female PhD Students, Post Docs, Senior Research Associates and Faculty; Organizer: oversaw undergraduate and MA department TA and grader hiring; Organizer: Fall Macro MA Thesis Workshop; Reviewer: MAPSS-Econ applications (43 applications); Preceptor-Advisor: Econ MA students (5 advisees)

2021–2022 Committee Chair: Econ MA Thesis (6 papers); Committee Member: Econ MA Thesis (4 papers); Faculty Advisor: Blockchain Chicago RSO; Faculty Advisor: Undergraduate Women in Economics RSO; Feedback: MAPSS MA Thesis proposal presentations; Grader: Undergraduate Economics Placement Exam; Letters of Recommendation (9 students, some with multiple letters); Organizer: oversaw all undergraduate and MA department TA and grader hiring; Mentor: Professional Development Lunch for Female PhD Students, Post Docs, Senior Research Associates and Faculty; Reviewer: MAPSS-Econ applications (37 applications); Organizer: Fall Macro MA Thesis Workshop; Preceptor-Advisor: Econ MA students (5 advisees)

2020–2021 Faculty Advisor: Undergraduate Women in Economics RSO; Feedback: MAPSS MA Thesis proposal presentations; Grader: Undergraduate Economics Placement Exam; Judge: Oeconomica Metrics Game; Letters of Recommendation (14 students, some with multiple letters); Mentor: Professional Development Lunch for Female PhD Students, Post Docs, Senior Research Associates and Faculty; Organizer: oversaw all undergraduate and MA department TA and grader hiring; Preceptor-Advisor: Econ MA students (5 advisees); Reviewer: MAPSS-Econ applications (41 applications); Thesis 2nd Reader: MAPSS Econ MA Thesis (2 papers)

2019–2020 Committee Member: Macroeconomics Curriculum Review; Contributor: online-teaching methods and materials documents for the UChicago Economics department; Faculty Advisor: Undergraduate Women in Economics RSO; Grader: Undergraduate Economics Placement Exam; Feedback: Mock Presentation of UChicago Fed Challenge Team; Feedback: MAPSS MA Thesis proposal presentations; Interviewer, on-campus: Economics Lecturer, Rank 3 (4 candidates); Judge: Oeconomica Metrics Game (cancelled); Letters of Recommendation (13 students, some with multiple letters); Mentor: Professional Development Lunch for Female PhD Students, Post Docs, Senior Research Associates and Faculty; Organizer: created and implemented a method for all department TA and grader hiring; Preceptor-Advisor: Econ MA students (4 advisees); Presenter: Fall Family Weekend model class “Cryptocurrencies and Blockchain”; Post-address col-

loquium: Aims of Education; Reviewer: MAPSS-Econ applications (23 applicants); Thesis 2nd Reader: MAPSS Econ MA Thesis (5 papers); Trainer: how to use Gradescope for online assignments

2018–2019 Examiner: Undergraduate Economics Placement Exam; Feedback: Mock Presentation of UChicago Fed Challenge Team; Feedback: MAPSS MA Thesis proposal presentations; Interviewer, first round: Lecturer Recruitment (Economics Dept.); Interviewer, on-campus round: Lecturer Recruitment (Economics Dept.); Judge: Oeconomica Metrics Game (first round); Letters of Recommendation (3 students, some with multiple letters); Organizer: Summer Teaching Schedule; Preceptor-Advisor: Econ MA students (6 advisees); Reviewer: MAPSS-Econ applications (36 applicants); Thesis 2nd Reader: MAPSS Econ MA Thesis (5 papers)

Trinity University (selected service)

2012–2018 Founder and Organizer of the Trinity University Junior Faculty Research Workshop; Junior Faculty Recruitment (Economics Dept.); University Lecturers & Visiting Scholars Committee; Department Representative for various events

AREAS OF WORK

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- **Digital-asset markets and their data:** price differences across Bitcoin exchanges, and what Bitcoin prices reveal about exchange-rate manipulation and capital controls; whether market-efficiency findings depend on the price source, with a survey of the price sources used in earlier Bitcoin studies; global cryptoasset benchmarking and digital-mining energy studies; a master's course on cryptoasset markets; co-editor of an *Eastern Economic Journal* special issue on the economics of cryptoassets (Research and Publications; Teaching and Advising; Service to the Discipline).
- **Central-bank digital currency, stablecoin and monetary-policy analysis:** Federal Reserve Bank of Dallas publications, a Palgrave chapter, and presentations at De Nederlandsche Bank and the Federal Reserve Banks of Atlanta and Dallas; a Ph.D. course on the risks and benefits of digitized money; stablecoin panels for NABE Chicago and EconVue; co-editor of an *Eastern Economic Journal* special issue on the economics of cryptoassets, reviewing its CBDC submission; reviewing CBDC submissions for the Crypto Asset Lab conference selection committee (Research and Publications; Presentations; Current Positions; Teaching and Advising; Public Service; Service to the Discipline).
- **Monetary-instrument design and quantitative investment research:** a CPI-indexed asset-backed currency, designed, simulated, and tested; portfolio modeling, parameter search, and walk-forward validation; working papers on robust parameter search and on how universe construction creates the crypto cross-section (Methods and Skills; Selected Other Positions; Research and Publications).
- **Blockchain governance, tax and regulation, as economic analysis:** peer-reviewed governance and blockchain-forks articles; co-editor of an *Eastern Economic Journal* special issue with papers on Bitcoin forks and the environmental costs of proof of work (Research and Publications; Service to the Discipline).
- **Public explanation of digital money:** 10 audio interviews and 16 written or quoted media items; a CoinDesk opinion piece; talks and panels for non-academic audiences; Board Director for Monetary Policy and Executive Committee Advisor, NABE Chicago (Media; Selected Other Positions; Public Service).
- **International capital flows and growth:** when private savings, public debt and reserve flows move around a developing country's growth episode, and how the positive link between growth and capital outflows fades over twenty years (Research and Publications).
- **International trade:** how income inequality in the importing country changes the value, quantity and unit value of imported goods, in a model where firms price-discriminate by consumers' income (Research and Publications; Presentations).
- **Business cycles across developed and developing economies:** cycle lengths dated country by country, and the stylized facts re-estimated on them (Research and Publications).
- **Assessment and evaluation design:** a *Journal of Economic Education* article on assessment design, a multi-institution study of academic mindset in economics courses, to which I contributed review and suggestions (*International Review of Economics Education*), and instructor training (Research and Publications; Teaching; Presentations).
- **The economics profession:** changes in US academic economics department jobs, and a guide to the teaching-

focused academic job market; a *CSWEP News* piece on the job market for non-tenure-track academic economists (Research and Publications; Media).
- **Research review:** referee for 18 journals and funders; co-editor of an *Eastern Economic Journal* special issue; conference selection committees (Crypto Asset Lab, Tokenomics, International Trade and Finance Association, AEA CSWEP sessions) (Service to the Discipline).
- **Economics teaching:** courses from non-calculus principles to a Ph.D. course, across macroeconomics, microeconomics, international economics, policy analysis, and cryptoasset markets, at four institutions (Teaching and Advising).
- **AI-assisted research systems:** a multi-agent research system for replication, data pipelines, and writing; comparisons of models and review cascades, scored for accuracy, unsupported claims, and cost (Methods and Skills).

METHODS AND SKILLS

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Mathematical Economic Theory, Modeling, and Computation

Theory models

- Consumer and producer theory (utility maximization, the Slutsky decomposition, expected utility).
- General equilibrium of endowment and production economies.
- Oligopoly (Cournot, Bertrand, Stackelberg).
- Solow and neoclassical growth.
- Real business cycles (RBC, IRBC).
- Money (cash-in-advance, money-in-utility, OLG, Baumol–Tobin, seigniorage).
- Optimal taxation and government debt.
- The central-bank commitment problem.
- Heterogeneous-firm trade (Melitz).
- Ricardian, specific-factors, and Heckscher–Ohlin trade.
- Game theory (Nash equilibria, backward induction).
- Banking models of deposit-rate markdowns, including an OLG model of a monopolistic bank.
- Global games.
- Ramsey and Markov-perfect policy.
- Asset pricing (Markowitz portfolios, CAPM, C-CAPM).

Solution methods

- Lagrangian and Karush–Kuhn–Tucker conditions.
- Bellman equations and dynamic programming.
- Value-function iteration.
- Log-linearization.
- Calibration.
- Machine-checked proofs in Lean 4 (directed): 13 for the guarantees of a parameter-search method and 3 for the tests behind a stablecoin-flow risk signal.

Model building and computational simulation

- Built and simulated dynamic general equilibrium (DGE) models in MATLAB of CBDC adoption and its effects on interest rates and exchange rates, for a central-bank client.
- Solved and simulated a bilateral 55-country Melitz model with Pareto productivity, non-homothetic preferences, and income-based price discrimination, calibrating its resale constraint by grid search and minimum distance to elasticities estimated from 2011 HS6 US export data.
- Built a Bayesian-learning growth model of a small open economy in MATLAB: beliefs about the components of TFP updated by a Kalman filter, TFP components estimated by GMM, and the model solved by value-function iteration on belief grids.
- Programmed teaching models in MATLAB: dynamic general equilibrium, RBC and IRBC, and money models; policy models solved for steady states and transition paths.

Econometrics, Time-Series Analysis, and Data

Empirical economics

- Pooled and country-by-industry OLS with fixed effects and clustered errors.
- Eaton–Kortum gravity estimation of trade barriers.
- Event-time cross-sectional regressions around growth episodes.
- Christiano–Fitzgerald band-pass filtering on country-specific windows.
- Jarque–Bera, Box–Pierce portmanteau, runs, and variance-ratio tests of the random walk.
- Unit-root and cointegration tests (DF-GLS, ADF, KPSS, Johansen trace, Pesaran–Shin–Smith bounds), vector error-correction models, and Granger and Toda–Yamamoto causality tests.
- OLS and logit regressions with marginal effects and clustered errors.
- Newey–West t-statistics and Fama–MacBeth regressions.
- Lowess detrending.
- GMM, nonlinear least squares, and minimum-distance estimation.
- Bayesian learning with the Kalman filter.
- Causal-inference designs (difference-in-differences, instrumental variables, and synthetic control), including directing their use in many MA theses.

Data

- Product-level trade (UN Comtrade, with CEPII distance and language data).
- Macroeconomic series (Penn World Table, World Development Indicators, OECD, IMF, external-wealth positions of Lane and Milesi-Ferretti).
- Exchange-level and aggregated cryptoasset prices and perpetual-futures funding rates.
- FRED and BLS series.
- AEA Job Openings for Economists and Universal Academic Questionnaire data.
- Central-bank statistics.

Macroeconomic, trade, and labor-market studies

- Constructed perpetual-inventory capital stocks (Caselli–Feyrer initial stocks, Penn World Table 7.1), labor-augmenting TFP, and a split of Lane–Milesi-Ferretti net foreign assets into public debt (WDI), reserves (IMF IFS), and a private residual, for 36 developing countries.
- Implemented a business-cycle turning-point algorithm on quarterly real GDP: a simplified Harding–Pagan version of Bry–Boschan, with Spencer-curve and moving-average rounds and censoring rules.
- Dated growth accelerations with a modified Hausmann et al. rule (77 episodes in 36 countries) and ran event-time regressions of capital flows around them, in eight specifications.
- Cleaned HS6 product-level trade data (unit-value trims, products sold to fewer than five destinations dropped, Rauch and BEC classifications merged) and estimated Eaton–Kortum trade barriers.
- Compiled central-bank series by hand for a CBDC book chapter.
- With Christopher J. Roark, classified 9,600 Job Openings for Economists postings by keyword in Stata, and built a representative economics department from the AEA's Universal Academic Questionnaire tables, for "Seeking Teachers."

Cryptoasset-market studies

- Built a daily bitcoin exchange-rate panel from prices in 52 currencies, matched to official rates from the IMF, BIS, FRED, and a commercial database (33 currencies kept), and tested it with DF-GLS, ADF, Phillips–Perron, and KPSS unit-root tests, VAR and Granger-causality tests, and a VECM.
- Cleaned trade-level records into exchange-rate series and measured the clustering of five-minute price moves.
- Validated the bitcoin-based measures of "Bitcoin Reveals Exchange Rate Manipulation and Detects Capital Controls" against Chinn–Ito capital openness and the IMF's exchange-rate regime classifications.
- Assembled daily prices from 11 bitcoin exchanges with an outlier rule, hand-collected 22 characteristics of each exchange, and tested the law of one price with Johansen trace, PSS bounds, and Toda–Yamamoto tests, and with split-sample cointegration tests around a suspected structural break.
- Pulled daily price histories for about 38,000 series from an API in Python, hand-collected one day's trading volume for every pairing, on every exchange, of the 50 largest cryptocurrencies, and computed each currency's exposure in the global crypto market.
- Regressed search series on bitcoin price, trading volume, and active addresses, to test whether attention turns into buying.

- With Ricardo T. Fernholz and Christoffer Koch, contributed to a daily crypto market-capitalization history for a study of power laws in crypto market caps.
- Compiled Bitcoin forks from two public lists and assembled price and trade data for 17 chain splits.
- Compared bitcoin prices from three price-aggregation sites with prices from the five largest exchanges, and computed log returns for each series.

Quantitative Research and Strategy Validation

Quantitative investment methods

- Systematic strategy design.
- Backtesting.
- Multi-objective parameter optimization.
- Sobol and quasi-random search.
- Stationary block bootstrap.
- Walk-forward and holdout validation.
- Overfitting diagnostics (deflated Sharpe ratio, probability of backtest overfitting, White's reality check, Hansen's SPA test).
- Morris sensitivity screens.
- Regime-switching scenario generation and Monte Carlo stress tests.
- Point-in-time universe construction.
- Concentration, specialization, and diversity measures (inverse Herfindahl indices).

Portfolio and monetary-instrument design

- Designed, simulated, and tested a modular, CPI-indexed asset-backed currency with a selectable basket of backing assets (sole designer) on AWS compute, and developed an inflation-tracking portfolio model screening 4,000+ assets.
- Developed the inflation-tracking model's monthly recommendations from roughly 4,167 assets, with beta screens, comparisons of loss functions, rolling validation, deduplication by principal components, and selection over millions of candidate portfolios.
- Directed speed-ups that cut the model's selection step from six hours to five minutes and an estimated full run from 3.1 years to 4.8 days, and raised the portfolios generated from about 25,000 in a three-day run to about 300,000 in one day.
- Designed a monthly portfolio data pipeline: ETF, fund, and crypto data, screens, and CVXPY optimization against inflation, crypto market-cap, and S&P 500 targets.

Strategy research and robustness testing

- Developed parameter-search and robustness-testing methods (Sobol sampling, local-stability checks, walk-forward optimization) for investment-strategy research.
- Scoped and ran a strategy-family study: 37 base strategies (moving-average strategies and signal strategies, such as RSI mean reversion and breakouts) and 1,052 variants, with fees and taxes, and a go/no-go assessment for a bitcoin trend-following product.
- Researched a stablecoin-flow risk signal and directed its strategy harness, write-up, and deployment specification.
- Designed a nine-step parameter-sweep pipeline with a six-phase adaptive grid, Hansen's SPA test, and automated reports.
- Developed the Robust Search method: a 69-page methods document, a working package, a standalone parameter-selection engine benchmarked against one live product and three proposed strategies, and team training.
- Directed robustness tooling: pre-test gates, a look-ahead causality check, multiple-testing corrections (deflated Sharpe ratio, probability of backtest overfitting, White's reality check, Hansen's SPA), a Morris sensitivity screen, a parameter adviser, and a champion–challenger registry.
- Directed a scenario-stress engine: a hidden semi-Markov regime generator with Monte Carlo ensembles.
- Directed a synthetic tokenized-gold price history, a reconciliation of differences between backtests, and a walk-forward study of bitcoin–gold rotation, which found no selection skill.
- Directed attention-driven rotation backtests (page views, volume spikes), point-in-time, with dead coins kept.
- Contributed a robustness module and a cost model to the team's shared backtester, and designed a standardized PDF strategy report.

- Assessed factor-ablation tests for index methodology.

AI Research Systems, Evaluation, and Tools

AI and engineering

- LLM research pipelines.
- Multi-agent orchestration with Git-based handoffs.
- Source-grounded LLM assistants.
- Claude Code, Codex.
- AWS compute.
- Git.
- LaTeX.
- MATLAB, Dynare, Excel, and Stata.
- Lean 4 (directed machine-checked proofs).
- Python and other languages through LLM-directed development.

Research systems and infrastructure

- Designed and direct a multi-agent AI research system for replication, data pipelines, and writing.
- Built an AI-assisted research pipeline across 6,000+ academic papers, turning an initial 600 studies into testable investment ideas.
- Migrated research code to LLM-assisted development (Claude Code, Codex) and validated the rewrite by reproducing the prior baseline portfolio before adding features.
- Built LLM companion packages for three proposed products.
- Set up a consolidated, Git-based local workspace to avoid lock-in to a single IDE or model provider.

Evaluation

- Directed comparisons of Claude models (Haiku, Sonnet, Opus) and multi-model review cascades across several tasks, including document drafting, research code, and the agents' own self-testing, scored for accuracy, unsupported claims, and cost, and used separation between agents as a quality control.
- Directed a multi-agent audit of the team's code and testing results.
- Directed synthetic-user personas to test a natural-language strategy workflow (identifying potential strategies from user prompts, and which strategies may appeal); the workflow had little success at either.